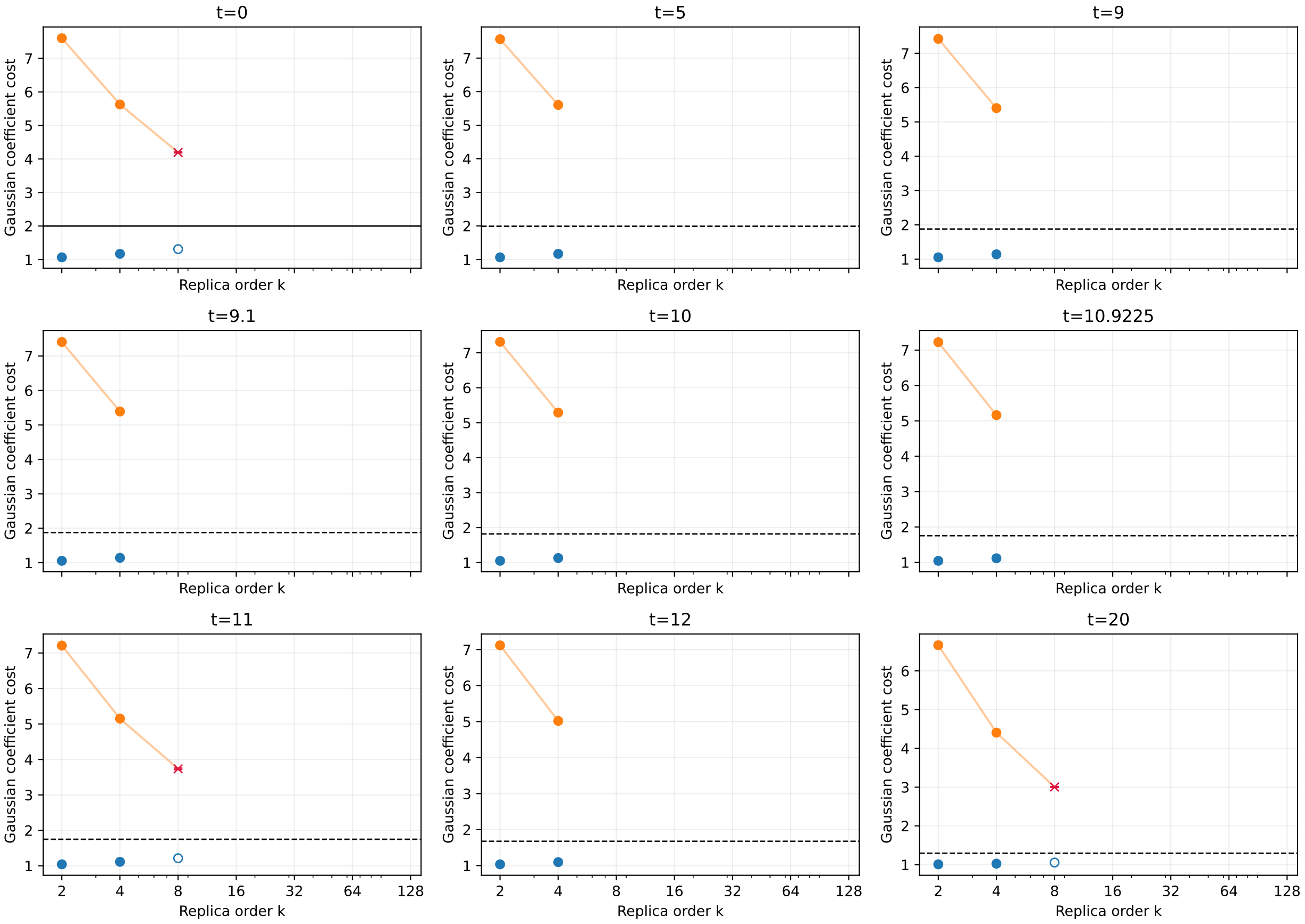Replica hierarchy: measured orders only; higher orders pending